Send Data Using LoRa® Technology with MKR WAN 1310

Learn how to setup a continuous stream of data between two devices using LoRa® technology.

Author · Karl Söderby Last revision · 02/28/2025

The Arduino MKR WAN 1310 is an excellent entry point to get started with low-powered, wide-area networks (LPWAN). In this tutorial, we will go through some of the core concepts such as LoRa® technology and set up communication using it.

We will create a basic sketch that will allow communication between two **MKR WAN 1310** boards. The communication will be established firstly through the radio module on the board, where we will also attach an antenna to each board.

Long Range, Low Power Networks

There are many different terms to be familiar with in the world of LoRa® technology, so let's go through some of them!

LoRa® Technology

LoRa® is short for long range modulation technique based on a technology called chirp spread spectrum (CSS). It is designed to carry out long-range transmissions with minimal power consumption. LoRa serves as the `lower layer` or `physical layer`, according to the **OSI model**. The physical layer is defined by hardware, signals and frequencies.

LoRa® uses different radio frequencies depending on where you are located in the world. The most common are Europe (868 MHz) and North America & Australia (915 MHz), but it differs from country to country. You can also read more about a country's unique radio frequency.

LoRa® is also often used to describe hardware devices supported by LoRa, e.g. modules or gateways. The Arduino MKR WAN 1310 has a LPWAN module called **Murata CMWX1ZZABZ**.

The Benefits of LoRa®

There are three distinct benefits of using LoRa® technology.

1. Price: LoRa® chips are quite inexpensive for its performance levels. The MKR WAN 1310 is one of the cheapest long-range connectivity boards produced. Also, operating under free licenses means, that's right, not paying for licenses!

2. Low power: Both hardware and software is designed to run on minimal energy, which boosts its value significantly. In present day and future, energy consumption is a large problem to tackle.

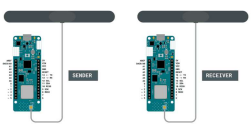
3. Long range: To transmit data over long distances brings more accessibility to more remote regions, and is ideal for agricultural, forestry and weather applications.

Hardware & Software Needed

- 2x Arduino MKR WAN 1310 (**link to store**)
- 2x Antenna (**link to store**)
- 2x Micro USB cable
- Arduino IDE (offline and online versions available)
- Arduino SAMD Board Package installed, **follow this link for instructions**
- `LoRa` library installed, see the **github repository**

Circuit

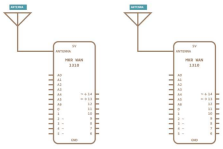
Follow the wiring diagram below to connect the antennas to the MKR WAN 1310 boards.



Sender & Receiver circuit with antenna.

Schematic

This is the schematic of our circuit.



Sender & Receiver schematic with antenna.

Let's Start

We are going to create two sketches: one for the sender, and one for the receiver. In this tutorial, we will simply focus on sending a simple message that can be read in the Serial Monitor. The sender will have a counter that increases each time the loop has run, and a delay of 5 seconds. Each time the loop runs, we will send a "packet" containing the message "hello" and the number of the counter variable. This is to ensure that we receive messages continuously.

To create the sender sketch, we will have to do the following steps:

- Initialize the **SPI** and **LoRa** libraries.
- Create a counter variable.
- Set the radio frequency to 868E6 (Europe) or 915E6 (North America).
- Begin a packet, print a message & counter variable

To create the receiver sketch, we will have to do the following steps:

- Initialize the **SPI** and **LoRa** libraries.
- Set the radio frequency to 868E6 (Europe) or 915E6 (North America).
- Create a function to parse incoming packet.
- Print the incoming messages.

Creating the Program

We are going to program two separate MKR WAN 1310's in this tutorial. We will start with the **sender** device and later on, we will create a sketch for the **receiver** device.

1. First, let's make sure we have the drivers installed. If we are using the Cloud Editor, we do not need to install anything. If we are using an offline editor, we need to install it manually. This can be done by navigating to **Tools > Board > Board Manager...**. Here we need to look for the **Arduino SAMD boards (32-bits Arm® Cortex®-M0+)** and install it.

2. Now we need to download the **LoRa** library from this repository, where you can install it by navigating to **Sketch > Include Library > Add .ZIP Library...** in the offline IDE.

The initialization is quick and easy: we will only include the **SPI** and **LoRa** libraries, and create a counter variable.

```
#include <SPI.h>
#include <LoRa.h>

int counter = 0;
```

In the `setup()` we will first begin serial communication, where we will use the command `while(!Serial);` to prevent the program from running until we open the Serial Monitor.

We will then initialize the **LoRa** library, where we will set the radio frequency to 868E6, which is used in Europe for LoRa® communication. If we are located in North America, we need to change this to 915E6.

```
void setup() {
  Serial.begin(9600);
  while (!Serial);

  Serial.println("LoRa

  if (!LoRa.begin(868E
    Serial.println("St
    while (1);
  }
}
```

In the `loop()` we start by printing "Sending packet" in the Serial Monitor + the value of `counter`. We then begin a packet

"hello" and the value of `counter`. This is done using the `LoRa.print()` function, then we broadcast it by using `LoRa.endPacket()`.

Finally, we add increase counter by 1 each time the loop has run, and a delay of 5 seconds, to limit the message rate.

```
void loop() {
  Serial.print("Sendin
  Serial.println(count

  // send packet
  LoRa.beginPacket();
  LoRa.print("hello ")
  LoRa.print(counter);
  LoRa.endPacket();

  counter++;

  delay(5000);
}
```

Programming the Receiver

The initialization and setup of the receiver is more or less identical to the **sender** sketch.

But inside the loop, we will not be creating any packets. Instead, we will listen to incoming ones. This is done by first using the command `int packetSize = LoRa.parsePacket();`, and then check for an incoming packet. If we receive one, it is parsed and printed in the Serial Monitor.

```
#include <SPI.h>
#include <LoRa.h>

void setup() {
  Serial.begin(9600);
  while (!Serial);

  Serial.println("LoR

  if (!LoRa.begin(868
    Serial.println("S
    while (1);
  }
}

void loop() {
  // try to parse pac
  int packetSize = Lo
  if (packetSize) {
    // received a pac
    Serial.print("Rec

    // read packet
    while (LoRa.avail
      Serial.print((c
    }

    // print RSSI of
```

{/ Here we link the full program from create /}

Complete Code

If you choose to skip the code building section, the complete code can be found below:

Sender Code

```
#include <SPI.h>
#include <LoRa.h>

int counter = 0;

void setup() {
  Serial.begin(9600);
  while (!Serial);

  Serial.println("LoR

  if (!LoRa.begin(868
    Serial.println("S
    while (1);
  }
}

void loop() {
  Serial.print("Sendi
  Serial.println(coun

  // send packet
  LoRa.beginPacket();
  LoRa.print("hello "
  LoRa.print(counter)
  LoRa.endPacket();

  counter++;
```

Receiver Code

```
#include <SPI.h>
#include <LoRa.h>

void setup() {
  Serial.begin(9600);
  while (!Serial);

  Serial.println("LoR

  if (!LoRa.begin(868
    Serial.println("S
    while (1);
  }
}

void loop() {
  // try to parse pac
  int packetSize = Lo
  if (packetSize) {
    // received a pac
    Serial.print("Rec

    // read packet
    while (LoRa.avail
      Serial.print((c
    }

    // print RSSI of
```

Upload Sketch and Testing the Program

Once we are finished with the coding, we can upload the sketches to the different boards. Remember that we need to upload the **sender sketch to the sender board** and the **receiver sketch to the receiver board.** This will require us to switch between the ports listed in the board manager.

When we open the Serial Monitor,

choose the **port** for your board, and open the Serial Monitor. The output should now be:

```
LoRa Sender
Hello 0
Hello 1
Hello 2
Hello 3
```

Now let's change ports to the **receiver** device, and open the Serial Monitor. We should now see the following:

```
LoRa Receiver
Hello 0
Hello 1
Hello 2
Hello 3
```

If we are receiving

Troubleshoot

If the code is not working, there are some common issues we might need to troubleshoot:

- Antenna is not connected properly.
- The radio frequency is wrong. Remember, 868E6 for Europe and 915E6 for Australia & North America.
- We have not opened the Serial Monitor.
- We are using the same computer for both boards without a serial interfacing program.

Conclusion

In this tutorial, we have introduced some fundamental concepts around LoRa® technology, where we have setup a basic communication line between two boards using the LoRa® technology-based communication. With this basic framework, you can go on to combine this tutorial with sensors and other software libraries, so that you can create your own long-range, low-powered devices!

Trademark Acknowledgments

> **LoRa®** is a registered trademark of Semtech Corporation.

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